Arrays	**Description**
Move Zeroes (Chocolate Factory)	Move all zero-valued packets to the end while maintaining relative order.
Sort 0-1-2 Array	Sort an array containing only 0s, 1s, and 2s.
Count Inversions	Count pairs (i,j) such that i < j and arr[i] > arr[j].
Majority Element	Find element occurring more than
Kadane's Algorithm	Find maximum sum contiguous subarray.
Subarray Sum Equals K Pattern	Count subarrays whose sum equals a target.
Set-Based Deduplication	Remove duplicates while preserving insertion order.
Maximum Guests at a Party	You are given: arrival Times and departure time. Determine the maximum number of guests present inside the hall at any point. (Similiar to Maximum Overlapping Intervals)
Longest Subarray with Given Sum K	An energy company records daily power generation. Find the longest continuous period whose total generated units equal K.
Mathematics / Combinatorics	
Circular Seating Arrangement	Count seating arrangements around a circular table with given constraints.
Handshake Counting	Count total handshakes among
Book Exchange Derangement	Count permutations with no fixed positions.
Count Sundays	Calculate number of Sundays under given calendar conditions.
Prime Numbers Between 1 and N	Print or count all primes in a range.
Count Sundays	
Strings	
Character Frequency Equalization	Given a string, remove exactly one character so that all remaining characters occur the same number of times.
String → Matrix Construction	Convert a given string into an r × c matrix while satisfying specific mathematical constraints.
Pattern Matching Problems	Identify or transform patterns within strings.
KMP / Pattern Searching	Find occurrences of a pattern inside a text efficiently.
Fake Palindrome Substrings	A substring is called secure if its characters can be rearranged into a palindrome. Count all secure substrings.
Invalid Transactions	Available on Leetcode
Matrices	
Lower Triangular Matrix Conversion	Convert a matrix so that all elements above the main diagonal become zero.
Matrix Reshaping	Rearrange matrix elements into another valid matrix dimension.
Rotate Matrix	Rotate an N × N matrix by 90 degrees.
Matrix Pattern Problems	Generate or transform matrices following a specific pattern.
Dynamic Programming	
Subset Sum Count	Count the number of subsets whose sum equals a target value.
Partition Equal Subset Sum	Determine whether an array can be divided into two subsets having equal sum.
Target Sum	Assign + or – signs to numbers to achieve a target sum.
Coin Change	Find minimum coins or number of ways to form a value.
Longest Common Subsequence	Find the longest sequence appearing in the same order in two strings.
Longest Increasing Subsequence	Find the longest strictly increasing subsequence.
Knapsack Variants	Maximize value under weight constraints.
Minimum Cost Path	Reach destination with minimum cumulative cost.
Minimum Cost Path	Reach destination with least possible cost.
Graphs	
Dijkstra's Algorithm	
Detect Cycle in Graph	
Prim's Algorithm	Construct a Minimum Spanning Tree using greedy selection.
Kruskal's Algorithm	Build MST using edge sorting.
Fence Equal Height Problem	Divide elements into two groups of equal sum while maximizing common height.
Trees	
BST Implementation	Insert, delete, search, and traverse nodes in a Binary Search Tree.
Lowest Common Ancestor	Find the nearest common ancestor of two nodes.
Symmetric Binary Tree	Determine whether a tree is a mirror of itself.
Root-to-Leaf Path Problem	Find paths satisfying a given condition or sum.
Tree Traversal (inorder, preorder, postorder , level order	Asked in Interviews
Binary Search	
Median of Stream Or Median of two sorted arrays	Continuously maintain median while numbers arrive.
Odd Occurring Element	In a sorted array where all elements occur twice except one, find the odd-occurring element.
Recursion & Backtracking	
Generate All Permutations	Generate valid subsets/combinations satisfying constraints.
Book Exchange (Derangement)	Count ways such that nobody receives their own book.
Three Item Quantity Combinatorics	Count valid combinations under quantity constraints.